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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001
M.Sc Degree Examination - APRIL 2021

CH204T – ANALYTICAL CHEMISTRY

Time : 3 hrs **Max Marks :75**

SECTION – A (20X1=20)

Answer **ALL** Questions

I. Choose the correct answer

1. Which is the absolute error?
   a. $x_m - x_n$ b. $\frac{(x_m - x_n)}{x_t}$ c. $\frac{x_m}{x_t}$ x 100 d. None of the above

2. The number of significant figures in 6.023 x 10^{23} is
   a. 3 b. 4 c. 7 d. 2

3. The indicator used in the titration of iodine against sodium thiosulphate is
   a. KI b. K_2CrOH c. $K_3[Fe(CN)_6]$ d. Starch

4. The indicator methyl orange works in the pH range
   a. 3.2 - 4.4 b. 1.1 - 2.2 c. 8.2 - 10 d. 4.8 - 6.0

5. The simplest whole-number ratio of atoms in a compound is called the _____.
   a. Empirical formula b. Molecular formula c. Formula mass d. None of the above

6. What is the empirical formula of a compound containing 0.347 mole P to 1.031 mole Cl?
   a. PCl_3 b. PCl_5 c. P_2Cl_5 d. P_2Cl_6

7. What does a high resistance reading in both forward-and reverse-bias directions indicate?
   a) A good diode b) An open diode c) A shorted diode d) A defective ohmmeter

8. How many terminals does a diode have?
   a) 1 b) 2 c) 3 d) 4

9. Which of the following is known as metal rectifier
   a. Selenium disc rectifier b. Copper oxide rectifier c. Gas tube diode d. All of the above

10. Molality is
    a. mols of solvent / kg of solvent b. mols of solute / kg of solute
    c. mols of solvent / kg of solute d. mols of solute / kg of solvent

II. Fill up the blanks

11. The square of standard deviations is called _____.

12. Phenolphthalein changes color in _____ solutions.

13. Law of normality is _____.

14. A mercury arc rectifier works on the principle of _____ by collision.

15. A circuit breaker is a _____ device,

III. Match the following

16. Standard deviation	a. resistor
17. Primary standard	b. loss of hydrogen
18. Whole-number ratio of atoms	c. σ
19. Thermistor	d. sodium carbonate
20. Oxidation	e. empirical formula

SECTION – B (10X2=20)

Answer any **TEN** Questions

21. Define precision.
22. What is average deviation?
23. What is confidence limit?
24. What is normality?
25. Define mole fraction.
26. What is molecular formula?
27. State Gay Lussac's law.
28. Write the law of constant composition.
29. What is the Operational amplifier?
30. Define rectifier.
31. What is absolute error?
32. Define molarity.

SECTION- C (5X7=35)

Answer any **FIVE** Questions

33. Discuss the determinate and indeterminate errors. How are their eliminated?
34. Explain the complexometric titrations.
35. Discuss the two theories of indicators.
36. Explain the determination of equivalent weights by chloride method.
37. a) Discuss the of back titrations. **(3)**
    b) Write the differences b/w primary standard and secondary standard. **(4)**
38. Write notes on i. Transformer ii. Capacitor.
39. Calculate the equivalent weight of the following.
    i. NaOH ii. H_2SO_4 iii. Na_2CO_3

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